

Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor

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1. INTRODUCTION

In recent years, the use of image sensors utilizing complementary metal oxide semiconductors (CMOS) have become a new trend in radiation detection due to factors such as their low cost, general availability, and their direct response to X-rays and gamma rays. This paper presents the estimation of dose rate using a process of extracting the predominant colors of an image taken from a CMOS image sensor while being irradiated, and analyzing their percentage in relation to the image as a whole.

In a project in which the development of a remotely controlled submersible robot (ROV) was carried out, a generic camera commonly used in surveillance devices or car maneuvering assistance systems was used as means of visual inspection. Said camera had a CMOS sensor, and it was subjected to tests under different dose rates of gamma radiation at the CDTN gamma irradiation laboratory.

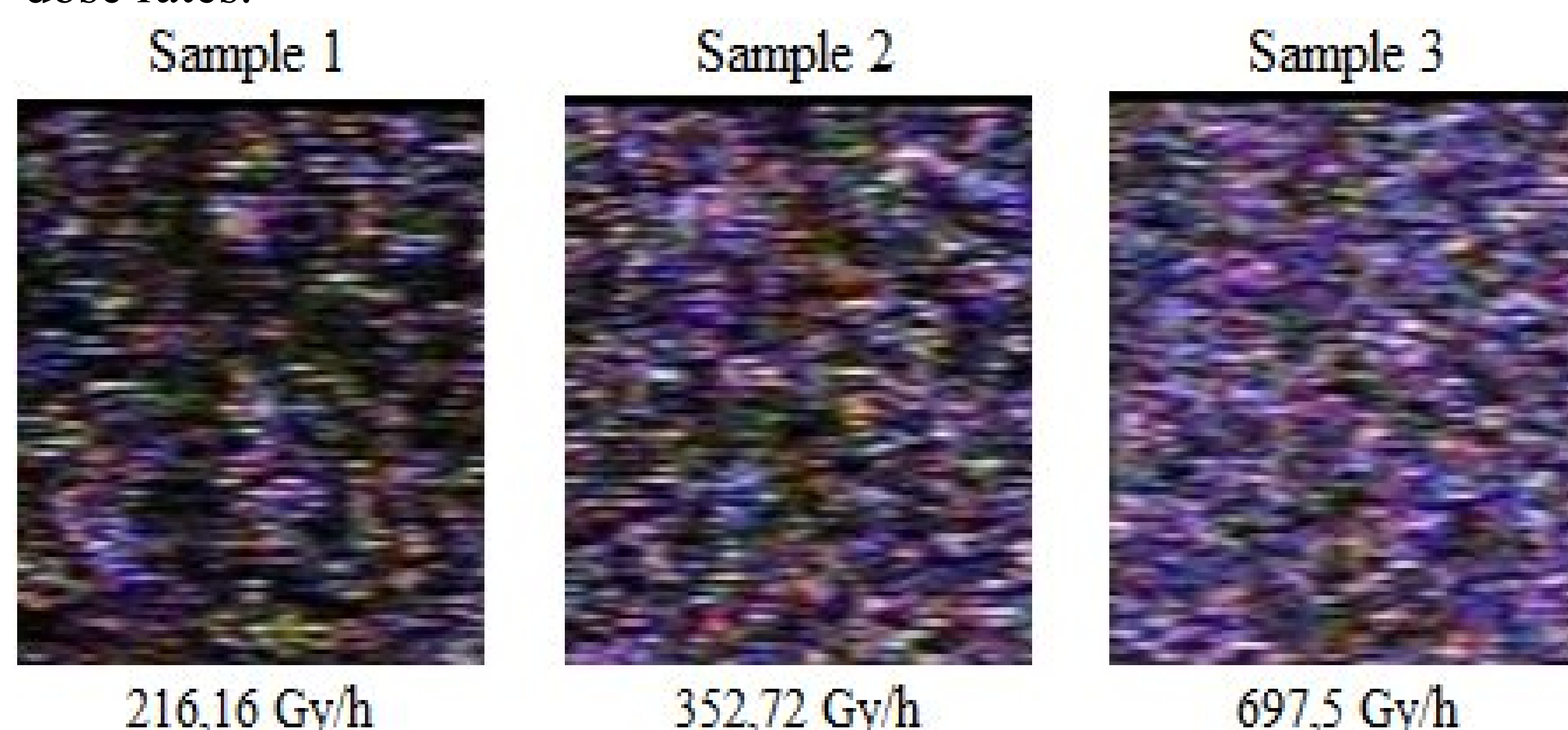
It is worth mentioning that there are also applications that can be installed on smartphones in order to carry out measurements of low dose rates using the image sensors installed on them.

2. MATERIALS AND METHODS

Several tests were carried out under different dose rates, with the aim of carrying out resistance tests on the ROV's internal circuit which was equipped with the before mentioned camera. It is worth highlighting here the possible variation of 10% in the recorded dose.

Samples were taken from the images, specifically from a black square of 100 by 100 pixels on their corners. Image 1 shows the samples taken from the images, at three different dose rates.

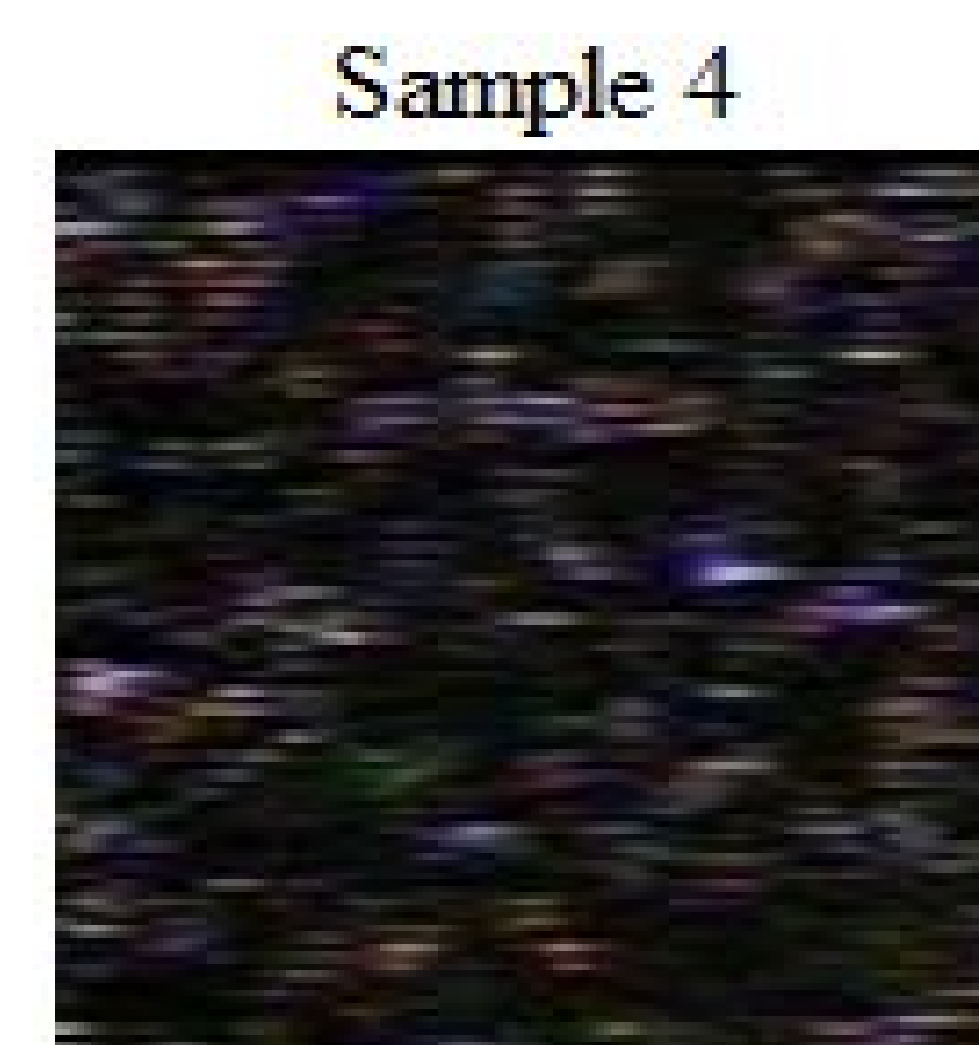
Figure 1: Three samples taken from three different images at different dose rates.



In all tests, the camera exhibited artifacts due to the passage of electrons in the circuit during irradiation, returning to the normal state as soon as irradiation stopped. A critical camera failure occurred at a total dose of approximately 250 Gy, 45 minutes after the tests began. This critical failure was repeated in all subsequent tests a few seconds after the start of exposure, indicating permanent damage to the camera's semiconductors, which prevented further data collection.

It is also worth highlighting the capture of an image in which the dose rate was unknown. Such an image was captured during the cover-up of the Cobalt-60 source, which resulted in such an unknown dose rate. This image generated sample number four.

Figure 2: Sample number 4, taken from an image of unknown dose rate.



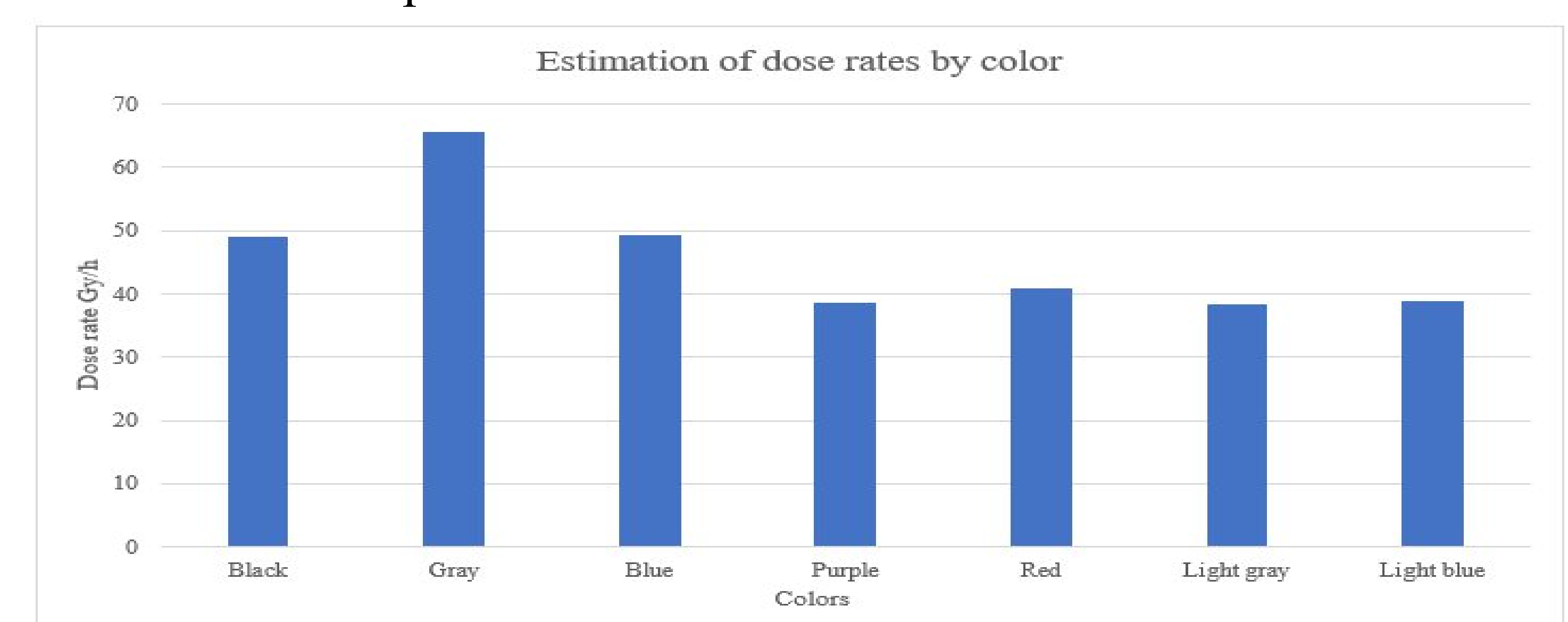
Subsequently, a tool was used to extract the predominant colors in the image. The PHP Image Color Extract tool extracts the most common colors from an image file. Color values are displayed in hexadecimal, along with their percentage in relation to the image.

Table 1: Percentage of each predominant color extracted from the samples.

Color (Code)	Control (%)	Sample 1 (%)	Sample 2 (%)	Sample 3 (%)	Sample 4 (%)
Black (#000000)	100	51,6	24,51	8,03	84,9
Gray (#646464)	0	25,25	38,78	47,36	9
Blue (#000064)	0	6,4	10,36	10,07	1,79
Purple (#640064)	0	6,18	11,8	15,45	1,23
Red (#640000)	0	4,14	3,4	1,17	1,54
Light gray (#c8c8c8)	0	1,71	3	3,24	0,2
Yellow (#646400)	0	1,51	1,03	0	0,7
Light blue (#6464c8)	0	1,2	3,48	9,09	0,09

3. RESULTS

Carrying out the calculations using the equations generated from the trend lines mentioned above, and considering the percentages of colors in sample four, it is possible to construct a graph which demonstrates the predicted dose rate for each of the colors.



4. CONCLUSIONS

The results of the dose rate estimate were also consistent with the appearance of the image presented in sample four, as it follows mathematical models that were generated from other images.

Subsequently, more tests should be carried out with better cameras, in tests with varying dose rates and of shorter duration, in order to preserve the integrity of the camera for more comprehensive data collection and creation of more accurate mathematical models. An algorithm can also be developed in order to process and analyze this data automatically.

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